

CuraAid

(Smart Health Care)



Author

Aneela Rafi

BCS231003

Laiba Riasat

BSE231001

Kashif Rehman

BSEB251002

Supervisor

Mam Maryam Sohail

Lecturer

Department of Computer Science Faculty of basic & Applied

Science M.Y University

Islamabad

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Author

Aneela Rafi

BCS231003

Laiba Riasat

BSE231001

Kashif Rehman

BSEB251002

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**Department of Computer Science Faculty of basic & Applied
Science M.Y University
Islamabad**

Final Approval

Thesis Titled: CuraAid

By

Aneela Rafi

BCS231003

Laiba Riasat

BSE231001

Kashif Rehman

BSEB251002

has been approved for
MY university, Islamabad

Thesis Supervisor:

Mam Maryam Sohail

Lecturer, Department of Computer Science

Thesis Supervisor Signature: _____

Head of Computer Science Department: _____

**Department of Computer Science Faculty of basic & Applied
Science M.Y University
Islamabad**

Abstract

A Smart and Secure Health Management Solution

The evolution of digital approaches has redefined health care in terms of enhanced accessibility, efficiency, and quality of medical services. On the other hand, there exists an array of existing systems by which appointment scheduling, patient records, consultations and prescriptions all are done separately on fragmented platform which contributes to communication gaps and administrative inefficiencies.

To this end, this project proposes curaAid, a one-stop web based holistic health care management portal that assimilates all key services in one place. It means that real-time interaction between patients, doctors and administrators is smooth. Patients can get preliminary healthcare advice about which doctor to visit according to the specialty, experience, stars and availability. The platform is equipped with online consultations including text, audio and video calls, appointment booking system, e-prescriptions, digital health records of patients, tracking cases for quick follow-ups or new visits as well as feedback on the consultation.

Interactive dashboards assist physicians in consultation hours, keeping patient records, e-prescriptions, monitoring treatment progress, analyzing statistics. administrative functions such as doctor verification, complaint management, user monitoring and system supervision. The pharmacy module allows users to upload prescription images to identify the medicine and prepare orders.

With the latest web technologies, curaAid is a safe, scalable and user-friendly environment. The purpose is to streamline administrative processes, enhance communication between stakeholders, improve the patient experience, and facilitate efficient healthcare delivery by integrating various healthcare services.

Undertaking

We certify that the research work titled “**CuraAid**” is our work. The work has not been presented elsewhere for assessment. Where material has been used from other sources, it has been properly acknowledged/referred to.

Signature: _____

Aneela Rafi

[BCS231003]

Signature: _____

Laiba Riasat

[BSE231001]

Signature: _____

Kashif Rehman

[BSEB251002]

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Chapter 1: Introduction

1.1 Introduction

The digital transformation of the healthcare sector has benefited both providers and patients. Modern systems ease the administrative burden and web-based platforms improve the speed and accuracy of digital medical records. Healthcare providers have easy and immediate access to medical records.

Even with these improvements, many healthcare facilities use disintegrated systems. Managing appointments, records, consultations, and prescriptions occurs in separate systems. This disintegration creates unnecessary administrative and communication burdens on staff, patients, and providers.

CuraAid is an all-inclusive integrated web-based system for the management of healthcare services. Patients can search for doctors, schedule consultations, book appointments, and access digital prescriptions and medical histories. Doctors have the capability to manage their schedules, conduct online consultations, and maintain medical records and the treatment of the patients. Healthcare administrators manage the system and have the capability to verify healthcare professionals.

The Initial Guidance Chatbot is a unique feature of the system. It guides patients to the appropriate healthcare professional based on their symptoms and medical needs, while considering the specialization, experience, ratings, and availability of the healthcare professional. The chatbot is not intended to diagnose or offer medical advice. Other features of the system are digital case management, consultation tracking, electronic prescriptions, and feedback and data analytics.

1.2 Background of the study

Healthcare providers around the globe are adopting new digital technologies to better serve their clients. technologies help to better deliver services, increase patient satisfaction and enhance clinical efficiency. Traditional healthcare systems depend on labor intensive methods. These include manual documentations, scheduling patient appointments, formulating and managing papers of prescriptions and maintaining scattered patient records. The tedious methods lead to several issues including data duplicity, communication lapses, and appointment scheduling conflicts.

Advent of modern internet technologies create a new era for healthcare systems. Integration of centralized digital systems modernizes the previously mentioned tedious methods. systems include online appointment scheduling, electronic medical records, virtual medical consultations, and e-prescriptions. These systems enhance service delivery and eliminate tedious methods.

Despite great innovations, most digital systems focus on only one or two healthcare services. Some systems focus on booking medical appointments only, while some focus on tele consults or medical record services. The fragmentation of digital systems forces the patient to utilize several digital systems to accomplish simple healthcare goals. It impacts user experience and increases administrative burden. Another major issue is the selection of desired healthcare professional. Many digital systems display a long list of healthcare professionals with little to no guidance.

Healthcare providers face operational issues in managing patient medical records, writing and tracking medical follow up prescriptions, and managing patient visits and ongoing cases. This problem is greatly magnified with increasing client volume and absence of an effective centralized system. In response to this, CuraAid has developed an all-in-one comprehensive system that combines intelligent doctor guidance, online consultations, appointment management, digital case tracking, e-prescriptions, medical histories, management, doctor verification, analytics and administrative oversight.

1.3 Problem Statements

Healthcare organizations deal with many challenges because many management systems break up the organization, and many administrative tasks are done manually. For example, many patients have to search for appropriate doctors and even have trouble scheduling their own appointments. Many patients also have trouble maintaining their medical records and searching for their history of past consultations. Patients often have to rely on various applications and manual tasks to complete many of these services, which causes delays and makes the process very inconvenient.

Choosing an appropriate healthcare provider can also be very challenging. Currently, many systems frame limited views of available healthcare providers. Some may even sort providers by specialty; however, many systems fail to provide advanced filtering tools to sort by experience, ratings, and even availability. This lack of appropriate provider selection may result in delays in many needed healthcare services.

Healthcare providers also have a very difficult time managing appointments, case histories of patients, and clinical activities. Maintaining case histories requires many system tools, which increases the administrative burden and workload for many healthcare employees. Managing many administrative tasks also requires special systems that are generally ineffective in current systems.

On top of all this, patients also have very limited views of the availability of doctors. Many patients also have very limited views of the progress of their cases and of their consultation history. This lack of transparent case tracking in healthcare makes management very inefficient.

Considering all these issues, there is an urgent need to develop a centralized web-based management system for the integration of patient management, doctor management, consultation services, scheduling services, electronic prescriptions and case tracking, doctor verification, management services, healthcare analytics, and healthcare guidance.

1.4 Proposed Solution

To address the deficiencies of current healthcare systems, CuraAid aims to provide multiple healthcare services in a single, centralized, web-based platform. An Initial Guidance Chatbot helps patients learn about healthcare providers. It takes into consideration a provider's specialty, years of experience, rating, and availability.

Patients using CuraAid have the option to perform a consultation via text, audio, or video. Based on the result of a consultation, a provider may recommend a course of treatment and prescribe medication. Physical treatment may be offered. The system provides patients with the ability to book appointments. Doctors are able to manage their schedule, give approval to appointments, and set their availability to control the number of appointments they will be doing for the day.

Digital patient case management fills in the gaps that appointment management systems leave. Doctors keep notes during a patient consultation along with their findings of a physical exam, the diagnosis, a treatment plan, and a list of follow up appointments. The notes for each appointment capture the patient's complete medical history and are retained by the system for future use. The system provides additional management through the generation of electronic prescriptions, case management, visit management, and a system for doctor's verification, complaints and feedback.

CuraAid's centralized system for healthcare focuses on improved accessibility and better coordination.

1.5 Objectives

The main aim of the CuraAid project is to design a secure, reliable and user-friendly web-based healthcare management system that will enhance communication, coordination and healthcare service delivery through digital technologies.

The specific objectives of the proposed system are:

- To develop a centralized healthcare management platform to integrate different healthcare services into one system.
- To help patients find the best healthcare professional through an Initial Guidance Chatbot based on specialization, experience, ratings and availability.
- To provide secure text, audio and video communication online consultation facilities.
- To facilitate appointments and help doctors manage their time and consultation schedules effectively.
- To keep up to date with all the details of the digital medical record including consultation history, examination notes, prescriptions and follow-up information.
- To produce electronic prescriptions that patients can safely access anytime they want.
- To provide complete case tracking to include ongoing, completion and revisit cases.
- facilitating the monitoring of patient progress by health professionals via structured digital case management.
- We offer analytical dashboards that allow patients, doctors and administrators to monitor healthcare activities and system performance.
- To create a secure doctor verification process. So that only the verified healthcare providers are able to provide medical consultation through the platform.

- To improve the quality of healthcare services by integrating patient feedback, complaint management and administrative monitoring.
- To develop a healthcare platform which is responsive, secure and scalable and can support future enhancements and additional healthcare services.

1.6 Scope of the Project

The scope of CuraAid is focused on providing an integrated digital healthcare environment where patients, healthcare professionals and administrators manage services efficiently through a centralized web application. The system encompasses the following functional areas.

Patient Module

Patients can sign up and login into secure accounts to handle healthcare activities. They can update their personal information, get initial guidance, search for suitable doctors, book consultations, book appointments, access consultation history, review electronic prescriptions, track case progress, upload medical reports, submit feedback and request revisit consultations.

Doctor Module

Healthcare providers are able to manage profiles, specify availability for consultation, carry out online consultation, review patient medical history, maintain consultation history, generate electronic prescriptions, update treatment progress, track ongoing cases and analyze consultation statistics through interactive dashboards.

Administrator Module

The administrator is in charge of the entire platform, including doctor credential verification, healthcare professional approval or rejection, user activity tracking, complaint management, appointment oversight, system integrity, and administrative report generation.

Consultation and Appointment Management

The system allows for different ways to have a consultation, such as through text, audio, or video. Doctors may recommend physical appointments when necessary. Patients check on the status of their appointments while doctors handle scheduling and availability.

Medical Records and Case Management

CuraAid keeps safe electronic medical records that include notes from consultations, medical diagnoses, prescription details, uploaded reports, past treatments, and information about follow-up care. Each patient's case is carefully followed from the first meeting all the way to the end, making sure their care is consistent and well managed.

Electronic Prescription System

Doctors issue digital prescriptions permanently available within the patient's account. Patients review previous prescriptions during future consultations without relying on paper documents.

Analytics and Reporting

Graphical dashboards present healthcare statistics for different users. Doctors monitor patient demographics, consultation trends, and common medical concerns, while administrators evaluate overall platform performance and service utilization.

Doctor Verification and Complaint Management

Before becoming active, every doctor completes verification by submitting professional credentials and supporting documents. Administrators review these before granting access. Patients can report inappropriate behavior, allowing administrators to issue warnings, temporarily suspend, or permanently disable accounts.

Project Limitations

The existing system is only set up to handle web-based healthcare management services. Advanced features like tracking emergency ambulances, connecting with wearable medical devices, managing labs online, doing automatic medical diagnoses, and linking with full hospital information systems are not included right now and could be added in the future.

1.7 Summary

This chapter explained why CuraAid was created by showing the problems with traditional healthcare management systems. The problem statement pointed out the issues with separate healthcare services, poor communication, and unconnected medical record keeping. A complete solution was introduced called CuraAid, which is a web-based platform that brings together doctor advice, patient consultations, scheduling appointments, managing medical records digitally, writing electronic prescriptions, analyzing data, and overseeing administrative tasks all in one system.

The project's goals were shared to explain what results the solution was meant to achieve, and the project's boundaries were set to outline what the solution would do and what parts of the system would be included. This chapter sets the stage for the rest of the report. This chapter gives a thorough look at existing healthcare platforms and related research to examine current solutions, find their shortcomings, and explain why creating the CuraAid system is necessary.

Chapter 2: Literature Review & Logical development

2.1 Introduction

The rise of information technology has changed the healthcare industry and driven growth in this sector with solutions focusing on improving communication, patient care and healthcare management via digitization. As healthcare providers turn to digital solutions for administrative forms, electronic medical records or third-party apps for online consultations and improved access, web-based platforms are becoming increasingly popular in the industry.

Multiple healthcare management systems have been designed to fulfil various parts of the delivery process. Some only offer online appointment booking: others focus on telehealth, electronic health records, and digital-prescriptions or healthcare information. Although these systems have contributed to improving accessibility, most are designed to perform specific functions rather than offering a fully integrated management environment.

Patients often have to juggle navigating across various platforms to perform multiple tasks relating to healthcare. Likewise, healthcare practitioners are faced with the hassle of keeping records on patients, maintaining consultation records, monitoring treatment schedules and planning other appointments via disparate platforms. Administrative tasks related to doctor verification, complaints and monitoring also need dedicated solutions that many of the currently existing platforms do not provide in a comprehensive manner

This chapter looks at current healthcare platforms and related research on managing healthcare digitally. By looking closely at their features, advantages, and shortcomings, it finds out where more research is needed, which makes the development of CuraAid necessary.

2.2.1 Existing Healthcare Projects

A review of existing healthcare management platforms provides valuable insights into the strengths and weaknesses of current digital healthcare solutions. Studying these systems helps identify the features that have successfully improved healthcare delivery while also highlighting the limitations that require further improvement [6], [7], [8].

Project 1: Practo

Practo is one of the most widely used digital healthcare platforms that connects patients with healthcare professionals through an online environment. The platform allows patients to search for doctors based on specialization, location, consultation fees, and patient reviews. Users can schedule appointments, consult doctors online, and access healthcare information through a single platform [6].

Key Features

- Online doctor search
- Appointment booking
- Online consultations
- Digital health records
- Doctor reviews and ratings
- Health articles and information

Limitations

- Limited case tracking functionality.
- Patient treatment progress is not comprehensively managed.
- Administrative monitoring features are limited.
- Advanced patient analytics are not available for healthcare professionals.
- Consultation history management could be more comprehensive.

Project 2: Zocdoc

Zocdoc is an online healthcare platform that simplifies doctor discovery and appointment scheduling. Patients can search for doctors according to their specialty, insurance compatibility, location, and available appointment times [8].

Key Features

- Doctor search
- Appointment booking
- Insurance-based filtering
- Patient reviews
- Appointment reminders

Limitations

- Focuses mainly on appointment management.
- Limited electronic case management.
- Limited patient progress tracking.
- Does not provide comprehensive healthcare analytics.
- Administrative healthcare monitoring is minimal.

Project 3: WebMD

WebMD is a healthcare information platform that provides medical knowledge, disease information, medication details, symptom guidance, and general health awareness. It serves as an educational resource rather than a complete healthcare management platform [7].

Key Features

- Medical information
- Symptom guidance
- Drug information
- Health education
- Wellness articles

Limitations

- Does not support appointment booking.
- No integrated patient management.
- No consultation management.
- No digital case tracking.
- No doctor verification process.

Comparative Analysis

The analysis of existing healthcare platforms demonstrates that each system addresses a specific aspect of healthcare management. Practo emphasizes online consultations and appointments, Zocdoc focuses on doctor discovery and scheduling, while WebMD primarily provides health-related information [6], [7], [8]. Although these platforms successfully solve individual healthcare problems, none of them offers a fully integrated environment that combines intelligent doctor guidance, consultation management, appointment scheduling, comprehensive case tracking, electronic prescriptions, analytics, doctor verification, complaint management, and centralized healthcare administration within a single web-based platform.

These observations indicate the need for a more comprehensive healthcare management solution capable of integrating multiple healthcare services into one centralized system. The proposed CuraAid platform has been designed to address these limitations by providing a unified healthcare ecosystem that enhances communication, improves operational efficiency, and supports better healthcare management for patients, doctors, and administrators.

2.2.1 Review of Research Articles

We looked through recent research on digital healthcare management, telemedicine, electronic health records, and online healthcare services. The studies point out a lot of positives with digital healthcare, but they don't shy away from the technical and operational problems either. All this gives some solid direction for building the CuraAid platform.

Research Article 1

Title: Changes in Physician Electronic Health Record Use with the Expansion of Telemedicine
Publication Year: 2023

Objective

The study investigated how the increasing adoption of telemedicine influences physicians' use of Electronic Health Record (EHR) systems during clinical practice [1].

Methodology

The researchers analyzed Electronic Health Record usage data from more than one thousand physicians working in multiple outpatient specialties. Physician activities before and after the expansion of telemedicine were compared [1].

Findings

The study concluded that telemedicine significantly increased physicians' interaction with Electronic Health Records, particularly for clinical documentation and patient record management. Digital healthcare platforms improved access to patient information but also increased documentation responsibilities for healthcare providers [1].

Limitation

Although the research demonstrated the importance of EHR integration, it mainly focused on physician workload rather than providing a complete healthcare management solution that integrates appointments, consultation management, patient case tracking, and administrative monitoring [1].

Research Article 2

Title: Impact of Asynchronous Electronic Communication Based Visits on Clinical Outcomes and Health Care Delivery

Publication Year: 2021

Objective

The purpose of this systematic review was to evaluate the effectiveness of electronic consultations in improving healthcare quality, reducing costs, and supporting patient management [2].

Methodology

The authors systematically reviewed multiple published studies related to electronic consultations across different healthcare environments and compared patient outcomes with conventional clinical visits [2].

Findings

The study reported that online consultations produced clinical outcomes comparable to face-to-face consultations for many healthcare conditions. Electronic consultations also reduced healthcare costs and improved convenience for patients while maintaining acceptable quality of care [2].

Limitation

The researchers observed inconsistent results across different medical conditions and recommended additional studies to improve healthcare quality, consultation management, and implementation strategies [2].

Research Article 3

Title: Differences in Physician Electronic Health Record Use by Telemedicine Intensity

Publication Year: 2024

Objective

This research evaluated the relationship between telemedicine usage intensity and physician interaction with Electronic Health Record systems [3].

Methodology

The study analyzed healthcare professionals from multiple academic medical centers by comparing Electronic Health Record usage across different consultation methods [3].

Findings

The results indicated that physicians providing more telemedicine consultations spent additional time maintaining patient documentation and managing digital healthcare records. The study emphasized that efficient healthcare platforms require optimized workflows to reduce administrative burden [3].

Limitation

The research concentrated primarily on physician documentation activities and did not address broader healthcare management features such as doctor verification, appointment management, patient analytics, or integrated case tracking [3].

Research Article 4

Title: Machine Learning for Multimodal Electronic Health Records-Based Research: Challenges and Perspectives

Publication Year: 2021

Objective

The study reviewed the use of structured and unstructured Electronic Health Record data for improving healthcare information management and clinical decision support [4].

Methodology

The authors examined numerous published studies focusing on multimodal Electronic Health Records and evaluated different techniques used to manage healthcare information [4].

Findings

The review highlighted that centralized Electronic Health Records significantly improve healthcare data accessibility, continuity of care, and medical information management. Combining multiple sources of patient information provides a more complete understanding of patient health history [4].

Limitation

The study identified challenges related to data quality, interoperability, and integration between different healthcare systems, indicating the need for more unified healthcare management platforms [4].

Research Article 5

Title: Natural Language Processing in Electronic Health Records in Relation to Healthcare Decision-Making: A Systematic Review

Publication Year: 2023

Objective

The research examined how modern language processing techniques can improve the management and interpretation of Electronic Health Records for healthcare decision-making [5].

Methodology

A systematic review of more than one hundred published studies was conducted to evaluate different approaches for processing clinical documentation and healthcare records [5].

Findings

The study concluded that digital healthcare systems can benefit from improved organization of medical records, efficient information retrieval, and enhanced clinical documentation. Better management of patient records contributes to improved healthcare coordination and decision-making [5].

Limitation

The researchers reported challenges associated with data quality, incomplete medical records, and differences in healthcare documentation standards across institutions [5].

Overall Analysis of Research Articles

The reviewed literature demonstrates that digital healthcare technologies have significantly improved healthcare accessibility, patient communication, Electronic Health Record management, and online consultation services. Recent studies consistently emphasize the importance of centralized healthcare information systems, secure digital records, and integrated consultation platforms for improving healthcare efficiency.

However, the literature also reveals several unresolved challenges. Existing solutions frequently focus on individual healthcare services rather than providing a comprehensive healthcare ecosystem. Limited integration between consultation management, appointment scheduling, patient case tracking, doctor verification, healthcare analytics, and administrative monitoring continues to reduce operational efficiency.

These research findings support the development of CuraAid as an integrated web-based healthcare management platform. By combining intelligent doctor guidance, online consultation, appointment management, digital case tracking, electronic prescriptions, doctor verification, analytics, and centralized administration within a single platform, CuraAid aims to address many of the limitations identified in current healthcare research and existing digital healthcare systems.

2.3 Limitations of Existing Systems

The analysis of existing healthcare management platforms shows that digital healthcare technologies have improved healthcare accessibility through features such as online appointments, telemedicine, electronic health records, and digital prescriptions. However, several limitations continue to reduce the overall efficiency and effectiveness of these systems [1] - [5].

A major limitation is the lack of integration among healthcare services. Most platforms focus on individual functionalities, such as appointment booking or online consultations, requiring users to switch between multiple applications. Existing systems also provide limited support for intelligent doctor selection, as patients must manually compare doctors based on specialization, experience, ratings, availability, and consultation fees [1], [2], [6], [8].

Doctor availability management remains another challenge, with many platforms lacking dynamic scheduling and real-time availability updates, resulting in appointment conflicts. Similarly, patient medical records are often fragmented across different healthcare providers, making it difficult to access complete treatment histories. Comprehensive patient case tracking from diagnosis to treatment, follow-up, revisit, and case completion is also insufficiently supported [3], [4], [5].

Administrative functions, including doctor verification, complaint handling, performance monitoring, and professional accountability, are often limited, affecting healthcare quality and patient trust. In addition, many platforms provide only basic healthcare analytics, restricting data-driven decision-making and performance evaluation [3], [6], [8].

Patient engagement and communication are also limited, as many systems focus primarily on appointment scheduling without supporting continuous treatment monitoring, consultation history, digital case management, or revisit management. Although electronic prescriptions are widely available, they are often not fully integrated with patient medical records and future consultations [2], [4], [5].

These limitations demonstrate the need for a comprehensive web-based healthcare management platform that integrates appointment scheduling, intelligent doctor recommendations, consultations, patient case management, electronic prescriptions, healthcare analytics, doctor verification, complaint handling, and administrative monitoring within a single secure and user-friendly system [1], [3], [5].

2.4 Research Gap

The literature review shows that even though current healthcare management systems have made healthcare better with features like online booking, telemedicine, electronic records, and digital prescriptions, most of these platforms work on their own instead of being part of a connected healthcare system.

A big problem in healthcare is that there's no single online system that brings together all the tools needed to manage patients, doctors, appointments, consultations, medical records, prescriptions, data analysis, and office tasks in one place. This method of splitting tasks makes things less effective and forces users to use several different programs.

Another issue is that there's no smart way to choose the right doctor. Most platforms let you search for providers based just on their area of expertise or where they're located, but they don't take into account things like how long they've been practicing, how patients rate them, when they're available, or what kind of care someone really needs. Currently, many systems do not fully support managing all stages of a patient's medical case, such as initial consultation, diagnosis, treatment plans, follow-up care, and final case closure.

The literature also points out problems with managing doctors' availability on a daily basis, handling administrative tasks like checking doctors' credentials and dealing with complaints, and using dashboards to track how well healthcare services are performing and the quality of care provided.

To fix these gaps, the CuraAid system brings together important healthcare services on one platform. It provides an Initial Guidance Chatbot to help suggest doctors, online consultations through text, audio, and video, booking appointments, tracking patient cases digitally, sending electronic prescriptions, managing doctor availability, healthcare data analysis, verifying administrative details, handling patient complaints, and keeping all patient records in one place. This combined method is meant to make healthcare easier to access, run more smoothly, improve communication, and boost the overall quality of services.

2.5 Chapter Summary

This chapter looked at current healthcare management systems and recent studies about digital healthcare technologies. The analysis found that although current platforms have made healthcare more accessible by offering services like online doctor visits, booking appointments, and keeping digital health records, there are still some issues that need to be addressed.

The main issues are that healthcare services are not connected well, patients don't get enough support or information, there isn't enough help for managing their cases, there's not enough oversight in the administrative part, healthcare data isn't analyzed properly, and there's no complete system that brings all healthcare services together. These results show that there is a need for one place where all these things can happen together: managing patients and doctors, having online talks between patients and doctors, booking appointments, writing prescriptions electronically, keeping track of medical cases, using data to make decisions, and overseeing the office work, all in one website.

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